

Daniel Dunmore

Curriculum Vitae

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Education

2024 – present	Ph.D. Student in Pure Mathematics (University of New South Wales) Supervisors: Assoc. Prof. Pinhas Grossman, Dr. Anna Romanov, Dr. Arnaud Brothier Topic: Classification of Non-Unitary Near-Group Fusion Categories
2019 – 2023	Bachelor of Advanced Science (Honours) (University of New South Wales) Majors: Pure Mathematics, Advanced Physics Cumulative WAM: 80 Honours Supervisor: Dr. Arnaud Brothier Honours Topic: From Subfactors to Richard Thompson's Groups and their Generalizations Honours WAM: 85 (First Class Honours)

Research Interests

My current research interests include

- categorification and categorical representation theory, such as with respect to the category of Soergel bimodules and its module categories;
- planar algebras, subfactor theory and fusion categories;
- group theory and representation theory, primarily with respect to infinite simple groups such as Thompson-like groups and their unitary representations;
- topological quantum field theory and low-dimensional topology.

Publications

Cifuentes, J. D., Tantt, T., Gilbert, W. et al., *Bounds to electron spin qubit variability for scalable CMOS architectures*, Nat. Commun. 4299.**15** (2024).

Mai, P. Y., Pereira, P. H., Alonso, L. A. et al., *Enhancement of Electric Drive in Silicon Quantum Dots with Electric Quadrupole Spin Resonance*, Phys. Rev. Res. **7**, 043134 (2025).

Talks

2025, Aug. 11th	Categorification and the Lusztig–Vogan Module UNSW Mathematics and Statistics Postgraduate Conference (Australia)
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2025, Apr. 7th	The Banach–Tarski Paradox and Amenability UNSW Masters/PhD Student Seminar (Australia)
2024, Dec. 5th	Contextualizing Categorical Representation Theory AAMS Student Conference (Australia)
2024, May 1st	Basics of Module Categories Tensor Categories and their Modules Learning Seminar (Australia) https://sites.google.com/view/tensorcategories/home
2023, Nov. 14th	From Subfactors to Richard Thompson’s Groups and their Generalizations UNSW Pure Mathematics Honours Seminar (Australia)
2022, Feb. 3rd	C^*-Algebras of Discrete Groups AMSIConnect (Australia) https://vrs.amsi.org.au/student-profile/daniel-dunmore/

Conferences

2024, Nov. 18th – 2024, Nov. 29th	Tensor Categories, Quantum Symmetries, and Mathematical Physics MATRIX (Australia) https://www.matrix-inst.org.au/events/tensor-categories-quantum-symmetries-and-mathematical-physics/
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Internships

2020, Sept. – 2021, June	ARC Centre of Excellence for Quantum Computation and Communication Technology
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Posters

Samuel, J., Dunmore, T., Dunmore, D., Saraiva, A., Coppersmith, S. N., *Bloch Sphere Model for Two Spin- $\frac{1}{2}$ Systems*, presented as part of the UNSW Talented Students Program (2020).

Technical Skills

- Proficient in C, Assembly, Magma, Sage, MATLAB/Octave, Python, Mathematica, C++, PHP, JavaScript and HTML5.
- Experience with COMSOL Multiphysics and LiveLink for MATLAB.